1 **(a)** 1 **[B1]**

(b) 9 **[B1]**

(c) x^3 **[B1]**

[Total Marks: 3]

2 **(a)** 5 **[B1]**

(b) $\frac{1}{36}$ **[B1]**

(c) t^8 **[B1]**

[Total Marks: 3]

3 **(a)** m^7 **[B1]**

(b) $\frac{1}{25}$ or 0.04 **[B1]**

(c) 7 **[B1]**

[Total Marks: 3]

4 **(a)** $\frac{1}{9}$ **[B1]**

(b) $\frac{2}{3}$ **[B1]**

(c) x^2 **[B1]**

(d) y^{12} **[B1]**

[Total Marks: 4]

5 **(a)** Adds powers: $-3 + 5 = 2$ or 2^2 **[M1]**

4 **[A1]**

(b) 9 **[B1]**

(c) 1 **[B1]**

[Total Marks: 4]

6 **(a)** $\frac{1}{8}$ **[B2]**

M1 for $(\sqrt[4]{16})^{-3}$ or $(2)^{-3}$ or $\frac{1}{16^{\frac{3}{4}}}$

(b) $4w^8y^{-2}$ or $\frac{4w^8}{y^2}$ **[B3]**

B1 for 4

B1 for w^8

B1 for y^{-2} or $\frac{1}{y^2}$

[Total Marks: 5]

7 **(a)** 27 or x^3 or y^6 seen **[M1]**

$27x^3y^6$ **[A1]**

(b) Writes 16 as 2^4 **[M1]**

$(2^4)^x = 2^{4x}$ **[M1]**

$4x = 10 \implies x = 2.5$ or $\frac{5}{2}$ **[A1]**

[Total Marks: 5]

8 **(a)** $16x^6y^2$ **[B2]**

If 0 scored, allow [M1] for a product containing two of 16, x^6, y^2

(b) $\frac{3a^4}{b^3}$ or $3a^4b^{-3}$ **[B2]**

If 0 scored, allow [M1] for a term containing two of 3, a^4, b^{-3}

(c) $\frac{1}{9}$ **[B2]**

If 0 scored, allow [M1] for evaluating the cube root to 3 or for showing $\frac{1}{27^{\frac{2}{3}}}$ or 9

[Total Marks: 6]

9 **(a)** **(i)** 16 **[B2]**

If 0 scored, M1 for $(\sqrt[3]{64})^2$ or 4^2 or $4096^{\frac{1}{3}}$ seen **[M1]**

(ii) $\frac{1}{16}$ or 0.0625 **[B1]**

(b) **(i)** $\frac{5a^4}{b^4}$ or $5a^4b^{-4}$ **[B2]**

If 0 scored, M1 for $\frac{15}{3} = 5$ or a^{6-2} or b^{3-7} **[M1]**

(ii) $2m^3n^{-1}$ or $\frac{2m^3}{n}$ **[B2]**

If 0 scored, M1 for $4^{\frac{1}{2}} = 2$ or $m^{6\times\frac{1}{2}}$ or $n^{-2\times\frac{1}{2}}$ **[M1]**

[Total Marks: 7]

10 **(a)** $16^{\frac{1}{4}} = 2$ or equivalent correct first step **[M1]**

$\frac{1}{8}$ **[A1]**

(b) 25 **[B1]**

x^6y^{-4} or $\frac{x^6}{y^4}$ **[B1]**

(c) Expresses terms with a common base, e.g. $4^x = 2^{2x}$ or $32 = 2^5$ **[M1]**

Sets up a correct linear equation from indices, e.g. $3x - 1 + 2x = 5$ **[M1]**

$x = 1.2$ or $x = \frac{6}{5}$ **[A1]**

[Total Marks: 7]

11 **(a)** Deals with negative index: $\frac{1}{125^{\frac{2}{3}}}$ **[M1]**

Takes cube root: $\sqrt[3]{125} = 5$ or 5^2 **[M1]**

$\frac{1}{25}$ or 0.04 **[A1]**

(b) Takes 4th root of numerator and denominator: $\frac{2a^2}{3b^3}$ **[M1]**

Cubes the expression: $\left(\frac{2a^2}{3b^3}\right)^3$ **[M1]**

$\frac{8a^6}{27b^9}$ **[A1]**

[Total Marks: 6]

12 **(a)** $\frac{4y^4}{5x^2}$ **[B3]**

B1 for reciprocal $\left(\frac{16y^8}{25x^4}\right)^{\frac{1}{2}}$ seen

B1 for $\sqrt{16} = 4$ and $\sqrt{25} = 5$

B1 for $(y^8)^{\frac{1}{2}} = y^4$ and $(x^4)^{\frac{1}{2}} = x^2$

(b) -6 **[B4]**

M1 for changing to base 2: $4 = 2^2$ and $8 = 2^3$

M1 for $2^{3x-1} \times 2^{2(x+2)} = 2^{3(x-3)}$

M1 for equating powers: $3x - 1 + 2x + 4 = 3x - 9$

[Total Marks: 7]

13 **(a)** Inverts fraction to remove negative power or correctly applies negative power to all terms, e.g. $\left(\frac{y^4}{16x^8}\right)^{\frac{1}{2}}$ or $\frac{16^{-\frac{1}{2}}x^{-4}}{y^{-2}}$ **[M1]**

Correctly evaluates the coefficient $16^{-\frac{1}{2}} = \frac{1}{4}$ or correctly halves powers **[M1]**

$\frac{y^2}{4x^4}$ or $0.25x^{-4}y^2$ **[A1]**

(b) Recognises that $4^x = 2^{2x}$ or substitutes $u = 4^x$ **[M1]**

Expresses 2^{2x+1} as 2×2^{2x} or $2(4^x)$ **[M1]**

Simplifies to a single exponential term, e.g. $-2^{2x} = -8$ or $u - 2u = -8$ **[M1]**

$x = 1.5$ or $x = \frac{3}{2}$ **[A1]**

[Total Marks: 7]

14 (a) $\frac{27y^6}{8x^9}$ or $3.375y^6x^{-9}$ **[B3]**

If 0 scored, allow [M1] for taking the reciprocal to remove the negative power: $\left(\frac{81y^8}{16x^{12}}\right)^{\frac{3}{4}}$

Allow a further [M1] for finding the 4th root of the expression: $\left(\frac{3y^2}{2x^3}\right)^3$

(b) 6 **[B3]**

If 0 scored, allow [M1] for expressing both sides with a common base 2, e.g. $(2^3)^x = (2^2)^{x+3}$

Allow a further [M1] for equating the powers to set up the linear equation $3x = 2(x + 3)$

(c) 1 **[B3]**

If 0 scored, allow [M1] for expressing all terms as powers of 5, e.g. $5^1 \times (5^2)^k = 5^3$

Allow a further [M1] for correctly forming the equation for the indices $1 + 2k = 3$

[Total Marks: 9]

15 (a) -1.5 or $-\frac{3}{2}$ **[B3]**

If 0 scored, M2 for $5^k = (5^3)^{-\frac{1}{2}}$ or $5^k = 5^{-\frac{3}{2}}$ **[M2]**

If 0 or M2 not scored, M1 for $\sqrt{125} = 125^{\frac{1}{2}}$ or $\frac{1}{125^{0.5}}$ or $125 = 5^3$ **[M1]**

(b) $\frac{27y^6}{8x^9}$ or $3.375y^6x^{-9}$ **[B3]**

If 0 scored, M2 for $\left(\frac{81y^8}{16x^{12}}\right)^{\frac{3}{4}}$ or $\frac{16^{-\frac{3}{4}}x^{-9}}{81^{-\frac{3}{4}}y^{-6}}$ **[M2]**

If 0 or M2 not scored, M1 for finding the 4th root: $\frac{2}{3}$, x^3, or y^2 correctly applied to at least one term **[M1]**

(c) 1.5 or $\frac{3}{2}$ **[B3]**

If 0 scored, M2 for $2x - 1 + 2x = 5$ or $4x - 1 = 5$ **[M2]**

If 0 or M2 not scored, M1 for writing with common base 2: $4^x = 2^{2x}$ or $32 = 2^5$ **[M1]**

[Total Marks: 9]